Sequences and series of functions

Math 163 • Prof. Chowdhury

The next page has 14 ideas as boxes, with 15 numbered arrows between them and nothing written on any arrow. An arrow is a relationship between two ideas, and not always an implication. Sometimes the first box gives you the second. Sometimes it doesn't, and there's a counterexample. And sometimes each gives the other.

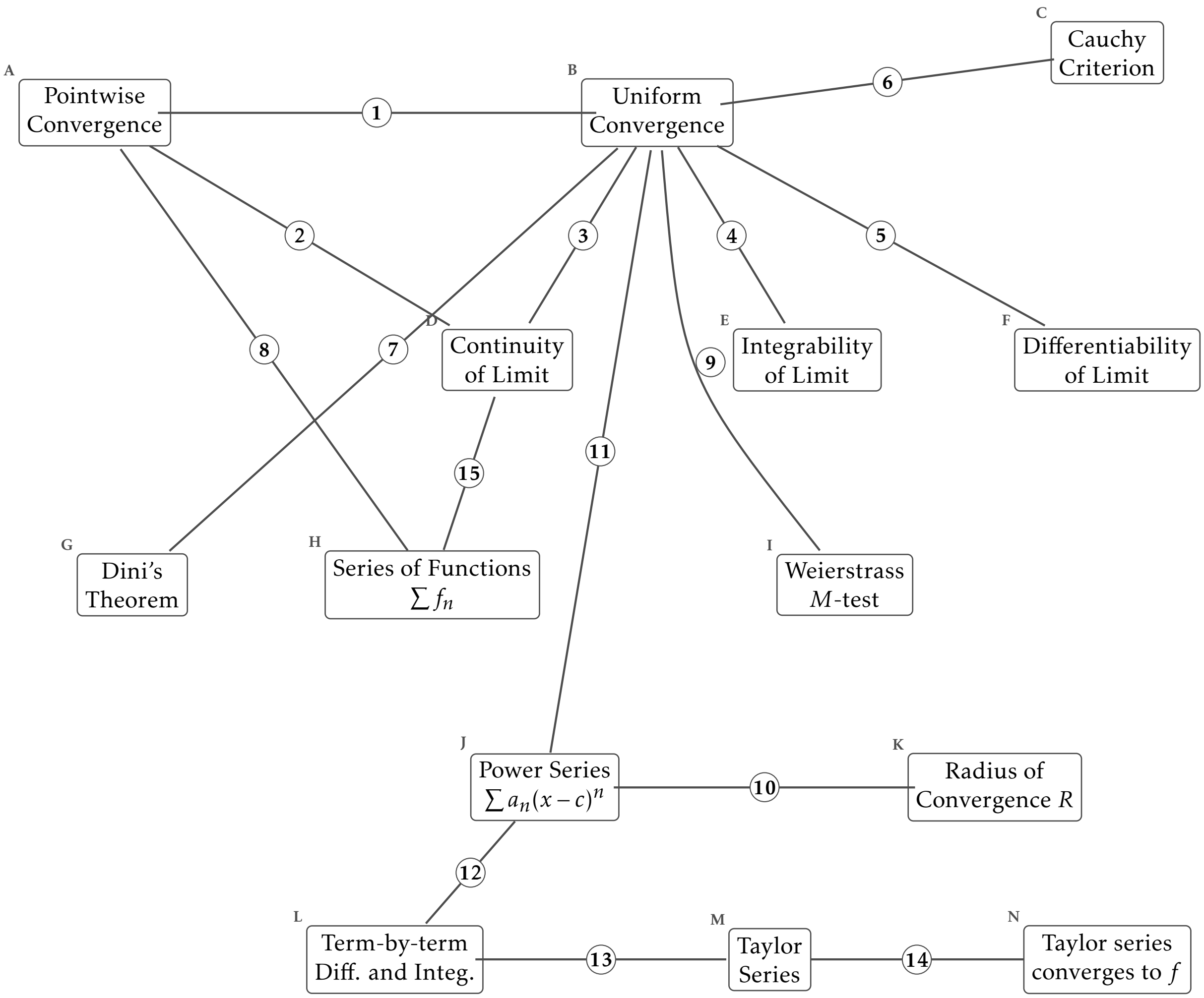
For each arrow, circle which of the three it is, and describe the relationship in one sentence. Name the theorem where there is one, and say what goes wrong where there isn't.

What each box means

- **A. Pointwise Convergence.** For each x and each $\varepsilon > 0$ there is an N depending on both, with $|f_n(x) - f(x)| < \varepsilon$ for $n \geq N$.
- **B. Uniform Convergence.** For each $\varepsilon > 0$ there is an N depending only on ε, with $|f_n(x) - f(x)| < \varepsilon$ for $n \geq N$ and every x at once.
- **C. Cauchy Criterion.** $\{f_n\}$ converges uniformly exactly when for each $\varepsilon > 0$ there is an N with $|f_m(x) - f_n(x)| < \varepsilon$ for all $m, n \geq N$ and all x.
- **D. Continuity of Limit.** Each f_n continuous and $f_n \to f$ uniformly gives f continuous.
- **E. Integrability of Limit.** Each f_n integrable and $f_n \to f$ uniformly gives f integrable, with $\lim \int f_n = \int \lim f_n$.
- **F. Differentiability of Limit.** $f_n \to f$ pointwise together with $f_n' \to g$ uniformly gives $f' = g$. It is the derivatives that have to converge uniformly.
- **G. Dini's Theorem.** $f_n \to f$ pointwise on $[a, b]$, monotonely, with every f_n and f continuous, gives $f_n \to f$ uniformly.
- **H. Series of Functions $\sum f_n$.** $\sum f_n$ converges when its partial sums $S_N(x) = \sum_{k \leq N} f_k(x)$ do. A series of functions is a sequence of functions.
- **I. Weierstrass M-test.** If $|f_n(x)| \leq M_n$ for every x and $\sum M_n < \infty$, then $\sum f_n$ converges uniformly and absolutely.
- **J. Power Series $\sum a_n(x - c)^n$.** A series of functions whose nth term is a monomial of degree n centred at c.
- **K. Radius of Convergence R.** $\sum a_n(x - c)^n$ converges absolutely for $|x - c| < R$ and diverges for $|x - c| > R$. The endpoints are checked one at a time.
- **L. Term-by-term Diff. and Integ..** Inside $(c - R, c + R)$ a power series can be differentiated and integrated term by term, and the result has the same radius R.
- **M. Taylor Series.** $\sum f^{(n)}(a)(x-a)^n/n!$. The coefficients are forced by differentiating the power series at its centre.
- **N. Taylor series converges to f.** The Taylor series converges to $f(x)$ exactly when the remainder $R_N(x) = f(x) - P_N(x)$ tends to zero.

Counterexamples and benchmarks

- x^n on $[0, 1]$ takes continuous functions to a discontinuous limit, pointwise.
- $\sqrt{x^2 + 1/n^2} \to |x|$ uniformly, and the limit is not differentiable at 0.
- $(1/n)\sin(n^2 x) \to 0$ uniformly while its derivatives diverge.
- $x/(1 + n^2 x^2) \to 0$ uniformly, and the derivative of the limit is not the limit of the derivatives at 0.
- e^{-1/x^2} extended by 0 is infinitely differentiable at 0 with every Taylor coefficient zero, so its Taylor series converges and not to it.



Every arrow is drawn the same way on purpose. Deciding the kind is half of each answer.

The arrows

1. gives / fails / both ways

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2. gives / fails / both ways

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3. gives / fails / both ways

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4. gives / fails / both ways

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5. gives / fails / both ways

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6. gives / fails / both ways

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7. gives / fails / both ways

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8. gives / fails / both ways

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9. gives / fails / both ways

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10. gives / fails / both ways

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11. gives / fails / both ways

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12. gives / fails / both ways

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13. gives / fails / both ways

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14. gives / fails / both ways

    ....

15. gives / fails / both ways

    ....

One last question

Which single arrow carries the most of this chapter? Say why in one sentence. If you pick one of the two that fail, say what you would have expected instead and what the counterexample costs you.

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